

CHAPTER 5

BASIC COMPUTER ORGANIZATION AND DESIGN



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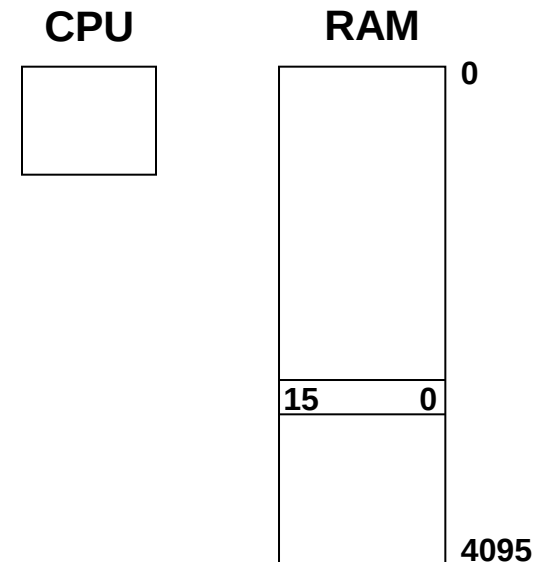
INTRODUCTION

- Every different processor type has its own design (different registers, buses, microoperations, machine instructions, etc)
- Modern processor is a very complex device
- It contains
 - Many registers
 - Multiple arithmetic units, for both integer and floating point calculations
 - The ability to pipeline several consecutive instructions to speed execution
 - Etc.
- However, to understand how processors work, we will start with a simplified processor model
- This is similar to what real processors were like ~25 years ago
- M. Morris Mano introduces a simple processor model he calls the *Basic Computer*
- We will use this to introduce processor organization and the relationship of the RTL model to the higher level computer processor



THE BASIC COMPUTER

- The Basic Computer has two components, a processor and memory
- The memory has 4096 words in it
 - $4096 = 2^{12}$, so it takes 12 bits to select a word in memory
- Each word is 16 bits long



INSTRUCTIONS

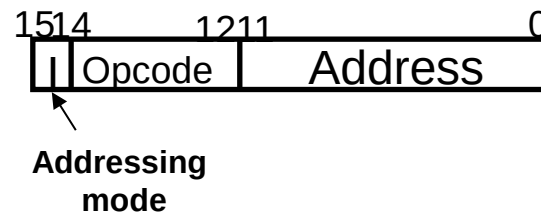
- Program
 - A sequence of (machine) instructions
- (Machine) Instruction
 - A group of bits that tell the computer to *perform a specific operation* (a sequence of micro-operation)
- The instructions of a program, along with any needed data are stored in memory
- The CPU reads the next instruction from memory
- It is placed in an *Instruction Register* (IR)
- Control circuitry in control unit then translates the instruction into the sequence of microoperations necessary to implement it



INSTRUCTION FORMAT

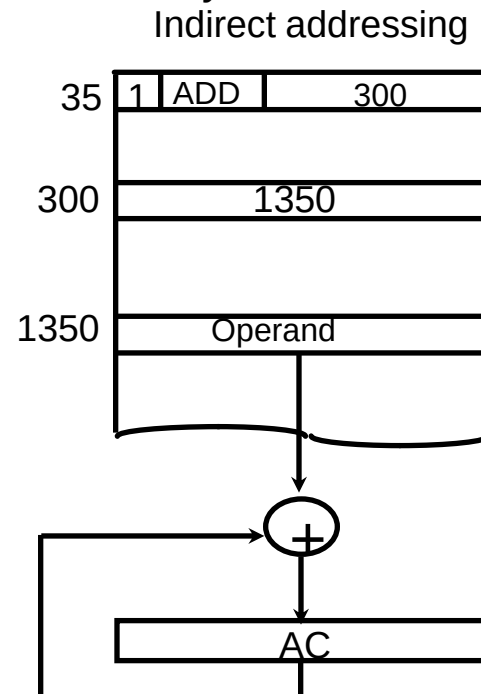
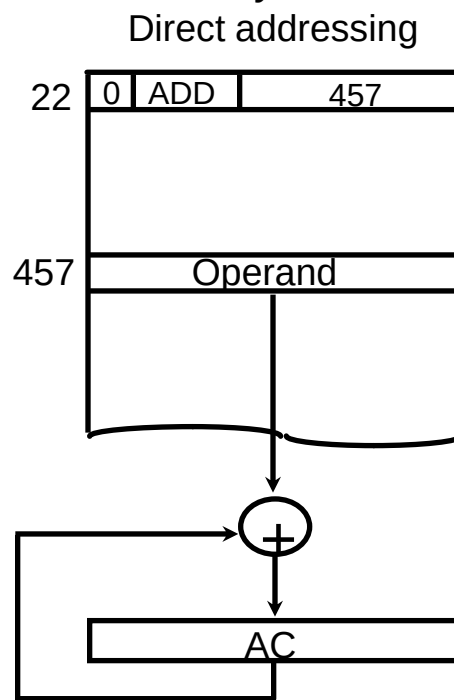
- A computer instruction is often divided into two parts
 - An *opcode* (Operation Code) that specifies the operation for that instruction
 - An *address* that specifies the registers and/or locations in memory to use for that operation
- In the Basic Computer, since the memory contains 4096 ($= 2^{12}$) words, we need 12 bit to specify which memory address this instruction will use
- In the Basic Computer, bit 15 of the instruction specifies the *addressing mode* (0: direct addressing, 1: indirect addressing)
- Since the memory words, and hence the instructions, are 16 bits long, that leaves 3 bits for the instruction's opcode

Instruction Format



ADDRESSING MODES

- The address field of an instruction can represent either
 - Direct address: the address in memory of the data to use (the address of the operand), or
 - Indirect address: the address in memory of the address in memory of the data to use



- Effective Address (EA)
 - The address, that can be directly used without modification to access an operand for a computation-type instruction, or as the target address for a branch-type instruction



PROCESSOR REGISTERS

- A processor has many registers to hold instructions, addresses, data, etc
- The processor has a register, the *Program Counter* (PC) that holds the memory address of the next instruction to get
 - Since the memory in the Basic Computer only has 4096 locations, the PC only needs 12 bits
- In a direct or indirect addressing, the processor needs to keep track of what locations in memory it is addressing: The *Address Register* (AR) is used for this
 - The AR is a 12 bit register in the Basic Computer
- When an operand is found, using either direct or indirect addressing, it is placed in the *Data Register* (DR). The processor then uses this value as data for its operation
- The Basic Computer has a single *general purpose register* – the *Accumulator* (AC)

